



Experimental and numerical study on vibrations and static deflection of a thin hyperelastic plate



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ABSTRACT

The hyperelastic behavior of a thin square silicone rubber plate has been investigated analytically, numerically and experimentally; the case of small-amplitude vibrations has been considered, as well as the case of large static deflection under aerostatic pressure. The Mooney-Rivlin hyperelastic model has been chosen to describe the material nonlinear elasticity. The material parameters have been identified by a fitting procedure on the results of a uniaxial traction test. For the analytical model, the equations of motion have been obtained by a unified energy approach, and geometrical nonlinearities are modeled according to the Novozhilov nonlinear shell theory. A numerical model has also been developed by using a commercial Finite-Element code. In the experiments, the silicone rubber plate has been fixed to a heavy metal frame; a certain in-plane pre-load, applied by stretching the plate, has been given in order to ensure a flatness of the surface. An experimental modal analysis has been conducted; results have been used to identify the applied in-plane loads by optimization procedure with two different models: a numerical and an analytical one. The first four experimental and numerical natural modes and frequencies are in good agreement with the experiments after the pre-load identification. The static deflection has been measured experimentally for different pressures. Results have been compared to those obtained by analytical and numerical models. The static deflections are also satisfactorily compared, up to a deflection 50 times larger than the plate thickness, corresponding to a 30 percent strain.

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1. Introduction

Thin walled structures composed of elastomeric materials are commonly found in engineering applications. The behavior of many vulcanized elastomers is closely described by the hyperelastic constitutive model [1,2]. This constitutive model is often applied to materials used in biomedical applications as well. The reduced thickness and stiffness of plates and shells composed by such materials give rise to large deformations under static operative loads. If dynamic loads are present as well, given the small thickness, large amplitude vibrations can be reached. The resulting problem is quite complex, as it

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features both geometric and physical-material nonlinearities. It is a common finding that the effect of physical nonlinearities become non-negligible for large static deflections only [3], while the effect of geometric nonlinearities on vibrations is reduced in a state of large initial static deflection [1,3–5].

Most relevant literature assumes as known *a priori* the deformed shape of the hyperelastic structures under study. A list of similar works is referenced in [6]. A second approach consists in approximating the deformed configuration with a truncated series of continuous functions that satisfy the geometric boundary conditions, as done in [5,7]. An alternative approach to the problem is the use of the Finite Element Method. Commercially available codes allow the nonlinear analysis of hyperelastic structures [8]. Einstein et al. [9] analyzed by FEM the dynamics of hyperelastic and viscoelastic membranes with vibration displacements in the order of the membrane's thickness. Verron et al. [10] investigated similarly the dynamic inflation of a Mooney–Rivlin spherical membrane. A last approach consists in the use of the Local Models Method [6]. Dynamic local models are built as polynomial expansions of the non-polynomial strain energy densities in the vicinity of a previously calculated static configuration. A system of ordinary differential equations with quadratic and cubic nonlinear terms in displacement is then used to approximate the behavior of the system. Breslavsky et al. [3] applied the Novozhilov nonlinear plate theory in the LMM study of a rectangular plate. They obtained the equations of motion of the system by an energetic approach and expanded the displacements in terms of a series of sine functions. The solution was found by means of the harmonic balance method [11]. Forced vibrations are investigated by the use of the software AUTO as well [12] and a softening behavior is predicted for a pressurized and deflected plate.

Several models are available for hyperelastic materials, namely the Neo-Hookean [7], Mooney–Rivlin, Yeoh, Ogden and Arruda-Boyce [5] models. Treloar [13] produced a collection of hyperelastic parameters for rubbers and biomedical materials. For most materials described by a hyperelastic constitutive model, the hypothesis of incompressibility holds true [1,2]. Gonçalves et al. [5,7] determined that the frequency-amplitude curves of pre-stretched incompressible membranes do not depend sensibly on the specific hyperelastic constitutive relationship chosen. Breslavsky [3] showed that the Mooney–Rivlin and the Neo-Hookean model yield similar results on thin square plates in static and dynamic cases. While these models can be applied to moderate strains (30 percent for rubbers and 8 percent for biomaterials), he also considers the Ogden model, which allows for higher strains. Ogden or Fung models should be used if strains are such that a sharp increase in stiffness for very high strains is reached.

While some experimental works on the characterization and identification of hyperelastic materials exists, literature featuring complete experimental tests on thin walled rubber-like structures is scarce. Alexander [14] studied the instability of neoprene spheres, Pamplona et al. [15,16] the free weight deformation of a liquid-filled membrane and the internal pressure inflation of a cylindrical shell.

The scarcity of experimental studies, on square inflated plates in particular, inspires the need of the present experimental work. While a static load is considered and vibration analysis is intended for characterization of the in-plane pre-load only, this investigation could constitute a preliminary step towards the future inclusion of nonlinear dynamics, as per the theoretical work by Amabili and his research group [3].

2. Hyperelastic material model

The silicone rubber used for the plate has a Shore (Type A) hardness of 70 and density of 1430 kg/m³, and it comes in a 1.5 mm thick roll. To characterize the material, a simplified traction test was performed applying increasing levels of weight to two strips of silicone rubber of cross-section 20 × 1.5 mm and 10 × 3 mm (the second one was obtained by using two layers of the silicone sheet), cut from the same raw piece as the tested plate. Two experimental stress–strain curves were obtained. Results are presented in Fig. 1.

In this study, we are treating silicone as an incompressible hyperelastic material. We use Mooney–Rivlin hyperelastic law to describe the nonlinear elasticity [1,2]. The elastic strain energy function for this law has the following form:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3), \quad (1)$$

where constants C_{10} and C_{01} are the material parameters of the model, and I_1 and I_2 are the first and the second invariants of the right Green–Lagrange strain tensor [2]. Three invariants of the strain tensor are expressed by formulas:

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2, \quad I_2 = \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_1^2 \lambda_3^2, \quad J = \lambda_1 \lambda_2 \lambda_3, \quad (2)$$

where λ_i are the principal stretches. For incompressible material, it is obtained $J = 1$ [2].

Since we intend to approximate the material's elasticity law with the expression (1) by using the uniaxial test data, we rewrite (2) for the specific case of uniaxial tension.

The principal Cauchy stresses are related to the partial derivatives of the strain energy function [1]:

$$\sigma_i = \lambda_i \frac{\partial W}{\partial \lambda_i}. \quad (3)$$

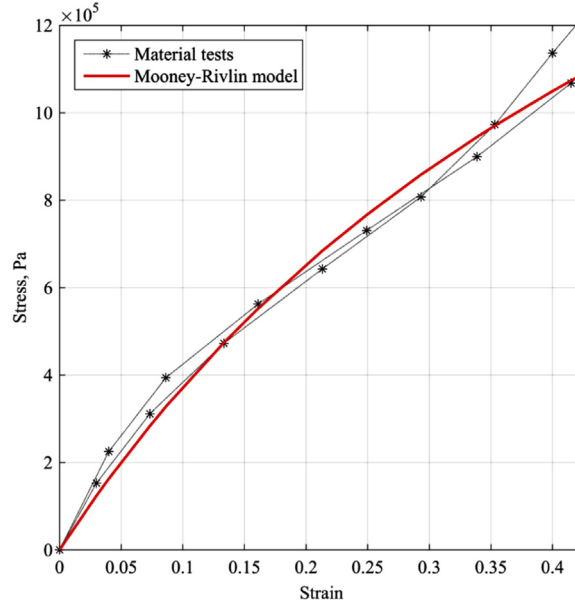


Fig. 1. Uni-axial traction tests on silicone strip, —*—, versus identified Mooney–Rivlin hyperelastic material model, —.

In case of uniaxial tension for incompressible material, we denote the principal stretch along the axis of tension $\lambda_1 = \lambda$, and the other two stretches are expressed as follows:

$$\lambda_3 = \lambda_2 = \sqrt{\frac{I}{\lambda}} = \frac{1}{\sqrt{\lambda}}. \quad (4)$$

From (1)–(4) the expression for the corresponding principal Cauchy stress can be easily calculated:

$$\sigma_{11} = 2C_{10} \left(\lambda^2 - \frac{1}{\lambda} \right) + 2C_{01} \left(\lambda - \frac{1}{\lambda^2} \right). \quad (5)$$

A tensile test output comes in the form of a set of data points, having strain and stress as coordinates. The tensile testing machine calculates strain and stress with respect to the initial dimensions of a sample (length and cross-section), and these values are thus called engineering strain ε^{ENG} and stress σ^{ENG} . That is why to compute the Mooney–Rivlin constants C_{10} and C_{01} from the material test, we first need to calculate the principal Cauchy stress and stretch from the uniaxial test data using the following equations [17]:

$$\lambda = 1 + \varepsilon^{\text{ENG}}, \quad \sigma_{11} = \sigma^{\text{ENG}} (1 + \varepsilon^{\text{ENG}}). \quad (6)$$

The model parameters C_{10} and C_{01} are further approximated using the least square procedure by fitting the model to the data of two uniaxial tensile tests carried out on silicone rubber samples. The stress–strain curve that corresponds to the best fit of Mooney–Rivlin model is presented in Fig. 1. Stability of the material model on the full range of strains has been verified by using the material evaluation tool available in the finite element software Abaqus [18]. The identified model parameters are: $C_{10} = 253216$, $C_{01} = 470900$.

3. Experiments

The experimental setup is composed by two subsystems: the square plate, made of silicone rubber, with its supporting frame, and the transducers and data acquisition system used to perform forced vibration testing and static pressure testing. In the following sections, these two subsystems are described. Hereafter, the experimental procedure and the relevant results are presented.

3.1. Silicone rubber plate and supporting frame

The vibrations of a 260×260 mm (square) silicone rubber plate, 1.5 mm thick, were investigated in this study. In reality, bigger square plates were cut from commercially available rolls, as part of the area had to be used to realize the desired

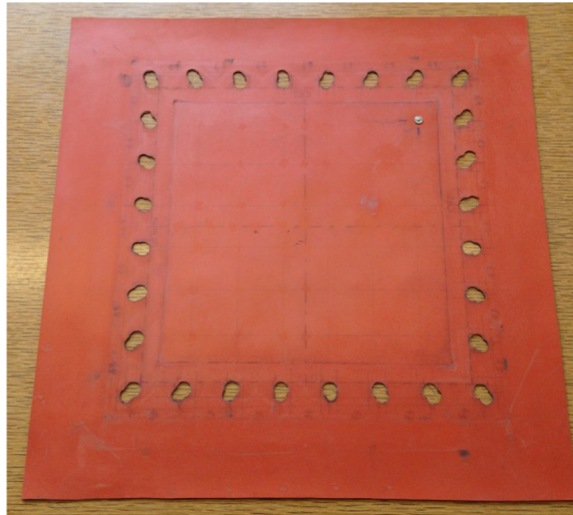


Fig. 2. Silicone plate with 28 punched holes for M12 bolts (the holes are deformed due to in-plane stress during tests).

boundary conditions. The silicone plate is shown in Fig. 2. The commercial catalog mentions a Shore type A hardness of 70 and density of 1430 kg/m^3 for the silicone rubber. While part of this study is intended to obtain the material characterization of these plates, it has to be noted that the Poisson's ratio was considered to be 0.5 (assuming incompressibility).

As silicone rubber is relatively soft and yielding, an in-plane pre-stretch was adopted to guarantee the flatness of the plate during vibration testing. In-plane tension was obtained cutting 28 holes in the plate and mounting it onto 28 perpendicular M12 fasteners, installed on a metal frame. The holes in the plate were uniformly distributed on a square perimeter, the side of which was 310 mm. The M12 rods, again equally-spaced, were centered onto a square perimeter, the side of which was 315 mm. The plates were therefore stretched to be mounted onto the threaded rods. The diameter of the holes is slightly bigger than the diameter of the bolts. The resulting in-plane tension can be described as an initial perpendicular displacement of 2.5 mm imposed to each side of the $310 \times 310 \text{ mm}$ square. As the distance between the holes is modest, the stretch can be considered uniform along the length of the sides of the square. This initial in-plane stretch should be perfectly symmetric; however, dynamic tests, described afterwards in this paper, reveal that the tension state is not perfectly symmetric. This can be due to the localized deformation of the silicone plates at the anchor points. Moreover, few tenths of a millimeter of stretch correspond to a large variation in the stress state. The square shape introduces an even higher incertitude about the strain distribution, as its corners constitute stress concentration factors. As mentioned above, the portion of the plates considered here is a square area, 260 mm of side, concentric with respect to the previously described offsets of holes and threaded rods. The 260 mm side of the vibrating square area is defined in the stretched side, and therefore corresponds to a smaller area in the relaxed state.

Fixed boundary conditions were realized pressing the silicone rubber plate between two heavy-weight metal frames matching square openings leaving the vibrating area free. The thickness and width and the rigidity of the frames guarantee a clamped condition. The resulting system is then connected rigidly to the ground. The two metal frames exert a pressure on the silicone plate by means of the same threaded elements used to keep the rubber into a pre-stretched state. The resulting friction force between metal and silicone is important in constituting the boundary condition; however, the two materials have a low friction coefficient and would require very a high torque on the threaded bolts. Being silicone nearly incompressible, higher torques would squeeze it between the two rigid frames, forcing it, eventually, to move into the free area. This complex transfer of material would buckle the originally flat free area and would appear as a pre-stretch reduction. Because of this, sandpaper was glued onto a portion of the contact area metal-silicone to increase the friction coefficient. Fig. 3 shows the silicone plate inside the metal frames.

The planar square area was maintained vertical during the experiments, with two sides parallel to the horizontal, since the effect of gravity was considered negligible with respect to the pre-stretch. A right-hand coordinate system was considered as having the x direction parallel to the horizontal and the z direction perpendicular to the plate and positive on the side of the plate opposite to the excitation system, described below. From now on, the observer will be considered standing on the positive side of the z axis. With this convention, the y axis will be vertical and pointing upwards. The origin of the coordinate system was chosen in correspondence with the bottom-left corner of the vibrating area, as described in accordance with the convention described. A transversal dynamic excitation (perpendicular to the plate) was positioned at $x=240 \text{ mm}$ and $y=220 \text{ mm}$. This location was chosen since modes with both odd and even symmetry had to be excited. The distance from the edges of the domain under study (the $260 \times 260 \text{ mm}$ square perimeter) was chosen as a trade-off. A reduced distance would correspond to a low energy transfer to the structure, being the edges subjected to a fixed boundary condition. A larger distance would instead give a wider unwanted interaction between the vibration of the structure and the excitation system. A punctual excitation was avoided as silicon is a soft, yielding and high friction material; it can present

localized modes of vibration, happening if the energy of the excitation remains limited to a small vibrating portion of the system. Therefore a rigid aluminum disk, 20 mm in diameter and negligible in weight, was centered on the chosen excitation point and glued by means of cyanoacrylic glue. The bond of aluminum and silicone through cyanoacrylic glue is weak; however it was deemed acceptable, the force amplitude involved in these experiments being low. Fig. 3 represents the coordinate system adopted and the position of the excitation.

For static deflections, an aerostatic load was considered. In order to apply air pressure, a closed air chamber had to be constituted on one side of the silicone plate. As described, two metal frames constitute fixed boundary conditions. Each frame is one inch (25.4 mm) thick and features one square opening (side 260 mm long). An air chamber 260 mm wide, 260 mm high and 25.4 mm deep was obtained closing the opening of one metal frame by means of a square aluminum flat plate. Airtightness was guaranteed pressing the aluminum square onto the frame with the use of eight adjustable C-clamps. A rubber gasket was interposed to avoid leaks. The aluminum plate was equipped with pneumatic components (gauge, valves and fittings), so that pressure could be increased, maintained and decreased upon connection to the compressed air service line. Fig. 4 illustrates the pneumatic system so implemented. It is to be noted that a close chamber of small volume can affect the vibrations of a flexible plate, independently on the value of the internal pressure (acoustic cavity). However, in this study pressure loading was not applied during dynamic tests. The back aluminum plate was actually removed during

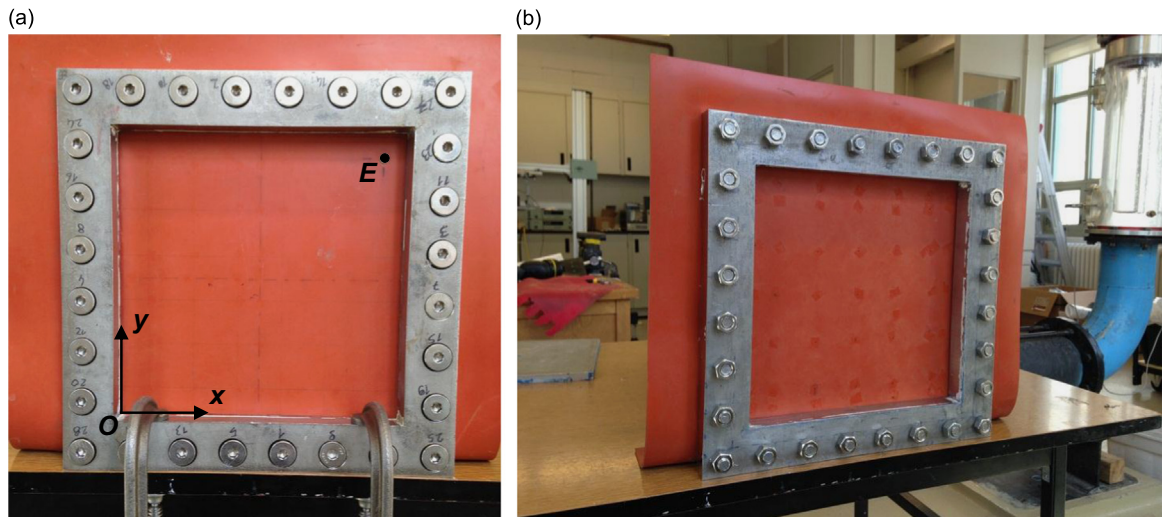


Fig. 3. Silicone plate inside the steel frame realizing fixed boundary conditions. (a) A planar coordinate system O, x, y is fixed onto the square area. The location E of the dynamic excitation is shown; (b) rear view.

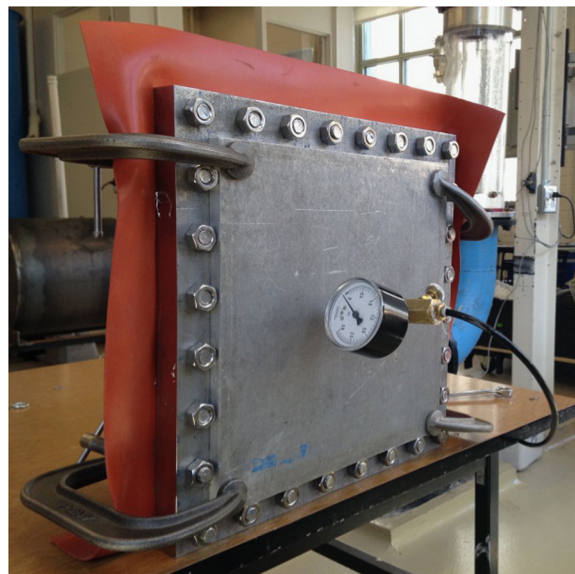


Fig. 4. Back plate complete with pneumatic components (pressure gauge and pneumatic inlet with needle valve) used for static pressure loading.

dynamic testing, in order to avoid the constitution of an acoustic chamber altogether. Furthermore, pressure loading was applied extremely slowly, so that the fluid-structure interaction does not have to be considered in the investigation.

3.2. Measurement apparatus

During dynamic testing, a Brüel & Kjær 8203 miniaturized force transducer has been used to measure the value of the excitation force. The transducer has been connected by a stinger to the electrodynamic exciter (shaker), a Brüel & Kjær 4810 powered by a Brüel & Kjær 2718 power amplifier. Fig. 5 shows an overall image of the excitation system applied to the silicone plate. The shape of natural modes of vibration has been obtained by means of scanning laser Doppler vibrometer (Polytec PSV-400). This technique allows the non-contact measurement of a multitude of points on the surface of the silicone plate, disposed on a fine grid. Non-contact measurement systems are favored also because they do not introduce unwanted added masses. Modal analysis made use of a dedicated Polytec OFV-5000 data acquisition system. Subsequently, the data have been transferred and processed by the LMS Test.Lab – modal analysis module software. Fig. 6 shows the apparatus used for modal analysis.

Static deflections under pressure loading have been obtained measuring the displacement of the center point of the plate. As expected, the center of the plate is the location of the maximum displacement in the inflated configuration. In order to measure the entity of its motion without any interference, a triangulation laser (Micro-Epsilon ILD 1402-200) was mounted on a tripod and aimed at the chosen point. This triangulation laser is capable of measuring relative static displacements with an accuracy of 0.1 mm. Fig. 7 shows the triangulation laser apparatus in the operative configuration.

3.3. Experimental results

The silicone plate under in-plane tension has been initially subjected to an experimental modal analysis, in order to characterize it and to obtain data necessary to identify with accuracy the in-plane tension. The vibration velocity of a set of points on the surface of the plate has been measured and processed by the LMS PolyMax algorithm [19]. As a result, the sum of the Frequency Response Functions has been obtained and it is represented in Fig. 8; the corresponding mode shape is

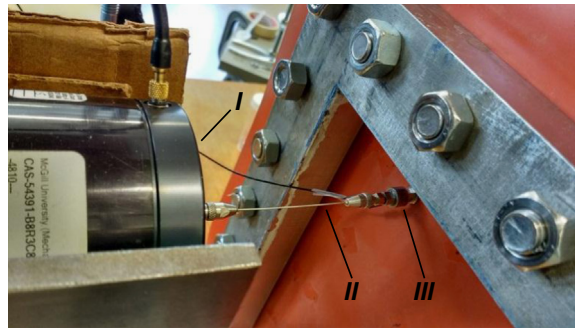


Fig. 5. Detail of the forced vibration excitation system applied to the structure: I electrodynamic shaker, II harmonic steel stinger, III force transducer connected to the vibrating plate by dedicated hardware.

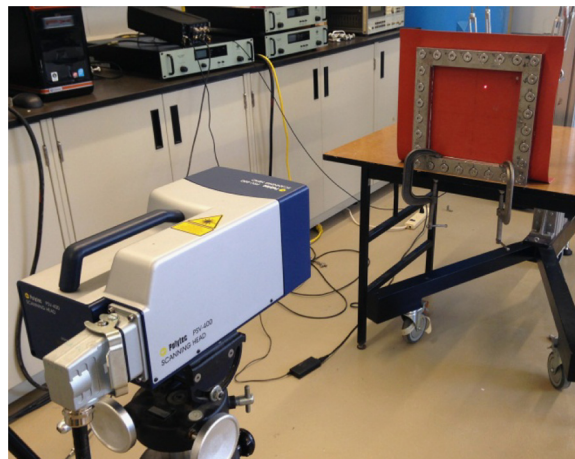


Fig. 6. Vibration measurement system (scanning laser Doppler vibrometer Polytec PSV-400) in operation onto the silicone rubber plate.

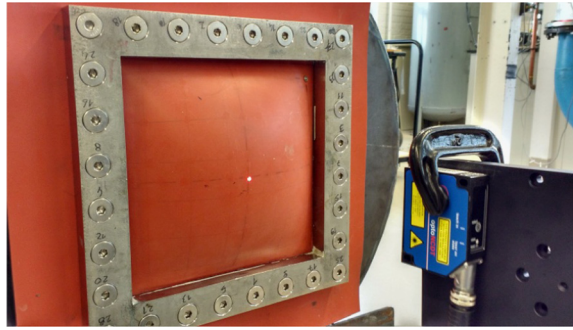


Fig. 7. Inflated configuration of the silicone plate under a 1 Psi pressure. A Micro-Epsilon triangulation laser sensor measures the deflection at the center point.

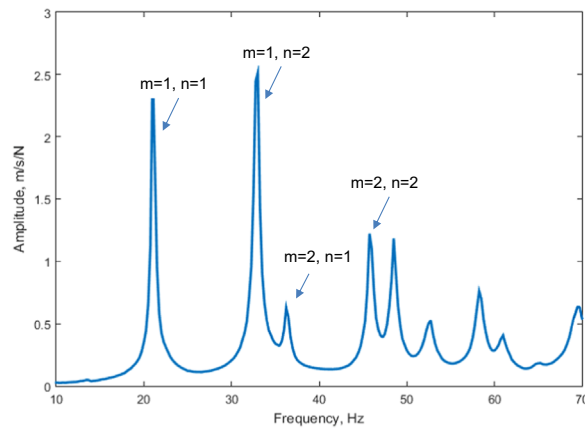


Fig. 8. Sum of the frequency response functions (FRFs) of the clamped silicone rubber plate.

Table 1

Experimental modal analysis results of the clamped silicone rubber plate. Natural frequencies and corresponding damping ratios.

Mode shape		Experimental results	
<i>m</i>	<i>n</i>	Natural freq. (Hz)	Damping (%)
1	1	21.0	0.53
2	1	33.0	0.56
1	2	36.3	0.82
2	2	45.7	0.61

indicated at each resonance peak. The natural modes are classified by the number of half-waves m and n in the x and y direction, respectively. The sequence of the first four modes, between 20 and 50 Hz, matches perfectly the one predicted by the numerical and analytical models, once the in-plane tension has been identified. It has to be noted that, for reasonable pre-load values, the initial pre-stretch does not change the sequence of the first modes. The experimental natural frequencies of the first four modes are given in Table 1, which also includes the modal damping ratios, calculated by the algorithm PolyMax. Modes (2,1) and (1,2) in Table 1 should occur at the same frequency, since the plate is square; however, the slightly different preload in the two in-plane directions determines a certain frequency shift.

Subsequently, the plate has been subjected to a pressure test. The pressure was increased in steps of 0.2 psi in the range 0–2.6 psi. The pressure variations have been performed very slowly and some minutes of settlement have been allowed after each pressure step had been reached, in order to minimize possible dynamic and viscous effects. Once the maximum desired value of pressure has been reached, the pressure has been decreased by steps according to a similar procedure. In spite of the slow process, hysteresis has been observed, as the curve obtained for increasing pressures does not follow the path of the one for decreasing pressures. The latter follows a higher deflection path. It has therefore been decided to subject the plate to various cycles of identical pressure loading. Not only each cycle has shown hysteresis, but the maximum deflection increased slowly with the number of cycles and a permanent deflection offset has also been accumulated after removing the pressure. These effects could not be attributed to a slippage in the boundary conditions, given the precautions described before, or to dynamics, given the extremely slow process. It is considered likely that semi-permanent

deformations were reached; however, these deformations would vanish in a time span of about 10 hours. As an ideal hyperelastic model is the object of the study, and since these phenomena are not quantitatively important, and as it is very common for polymeric materials to show a complex material behavior, it has been chosen to consider the first pressure loading as reference case for comparison to numerical simulations, as in this case no permanent deformation could have been accumulated. The experimental deflection curve for the first nine pressurization cycles is presented in Fig. 9.

4. Identification of the initial stretching of the plate by using experimental modal analysis data

During experimental set-up, the silicone plate has been stretched and assembled to the metal frame in a way to make this stretch as uniformly distributed along the sides as possible. However, to determine exactly the values of pre-stretch (i.e. the displacements along the edges), we introduce an optimization algorithm. The idea here is to match the experimentally measured natural frequencies of the silicone plate with the frequencies determined via analytical calculation and finite element method simulation. Thus, the optimization criterion is going to minimize the residual error function E_f constructed in the following form:

$$E_f = \sum_{k=1}^4 p_k \left(f_k^{\text{EXP}} - f_k^{\text{COM}} \right)^2. \quad (7)$$

In (7) f_k^{EXP} are the natural frequencies determined experimentally and f_k^{COM} are the frequencies computed from the model used; p_k is a weight coefficient voluntarily assigned to each of four modes considered. Typically the larger weight is assigned to the first mode, to give it a more significant impact with respect to the other modes.

The error function (7) has to be minimized by varying the initial dimensions of the plate, since the final dimension of the plate after stretching is known. The optimal set of parameters that minimize the error are the initial pre-stretching dimensions of the plate.

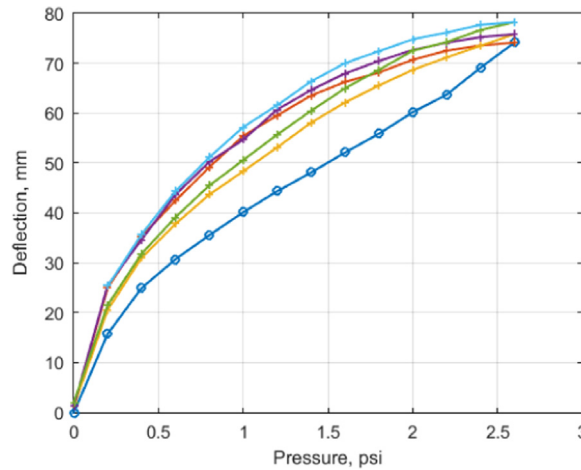


Fig. 9. Pressure vs deflection curves; all the cycles of loading and unloading combined. “o” – first loading cycle, “+” – all other consecutive loading cycles.

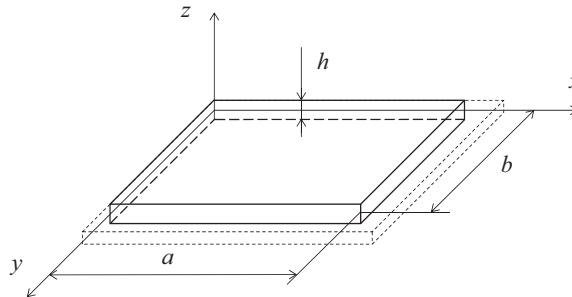


Fig. 10. Stretched plate and coordinate system.

4.1. Identification by using FEM

The pre-stretch phase is modeled as static analysis, where corresponding displacements are applied perpendicularly to the four edges of the plate. These displacements close the gap between the initial dimensions of the plate a and b and the known final dimension 0.26×0.26 m (as shown in Fig. 10). The identification code requires as an input the trial initial dimensions of the plate (a , b) and the increment of the two dimensions in the following steps, which determines the accuracy of the results. A square equally-spaced $N \times N$ grid of parameters is formed and passed on to the finite element (FE) software Abaqus and the simulation is launched N^2 times, which is the number of possible combinations of the two parameters.

The Mooney–Rivlin material model is used for the silicone rubber and the material parameters, obtained in Section 2, are $C_{10} = 253216$, $C_{01} = 470900$. The plate is clamped with the following boundary conditions:

$$u|_{\partial\Omega} = v|_{\partial\Omega} = w|_{\partial\Omega} = \left. \frac{\partial w}{\partial n} \right|_{\partial\Omega} = 0, \quad (8)$$

where $\partial\Omega$ is the plate's middle surface boundary and n is the normal to $\partial\Omega$ lying on the plate surface.

The parametric FE model of the plate is composed of four hundred 4-node conventional thin shell elements with six degrees of freedom at each node (displacements and rotations) and 5 integration points along the thickness [18].

After static stretching of the plate, the modal analysis is performed and the first four natural frequencies are extracted. Furthermore, as stated above, these frequencies are compared to the experimentally determined first four natural frequencies. The error (7) is calculated at every analysis and, once the cycle of N^2 simulations is completed, the minimal value of the residual error is the one associated to the optimal initial dimensions of the silicone plate. The best match of natural frequencies corresponds to initial dimensions of the plate $a = 0.2563$ m, $b = 0.2587$ m. The first four matched frequencies are presented in Table 2.

4.2. Identification by analytical procedure

We also used a different approach to the initial stretching identification. The residual error (7) is still computed, but with f_k^{COM} computed in approximate analytical form. We use the Novozhilov nonlinear shell theory [4] to describe the deformation of the plate; in fact, the plate is thin, but at the same time the in-plane displacements are large enough to require in-plane nonlinear terms in the model. The strain-displacement relations in rectangular coordinate are [4]:

$$\varepsilon_{11} = \frac{\partial u}{\partial x} + \frac{1}{2} \left(\left(\frac{\partial w}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right) - z \frac{\partial^2 w}{\partial x^2}, \quad (9a)$$

$$\varepsilon_{22} = \frac{\partial v}{\partial y} + \frac{1}{2} \left(\left(\frac{\partial w}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right) - z \frac{\partial^2 w}{\partial y^2}, \quad (9b)$$

$$\varepsilon_{12} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} - 2z \frac{\partial^2 w}{\partial x \partial y}, \quad (9c)$$

where u , v , w are the displacements of the plate's middle plane in x , y , z directions, respectively.

The potential energy of the plate has the form

$$\Pi = \iiint_V W dV, \quad (10)$$

where W is the strain energy density and V the plate volume. The strain energy density W takes a specific expression in case of Mooney–Rivlin material that can be found in [3]. The kinetic energy in case of flexural vibrations is given by [4]

$$T = \frac{\rho h}{2} \int_0^a \int_0^b \dot{w}^2 dx dy, \quad (11)$$

Table 2

Comparison of natural frequencies.

Mode shape		Experimentally measured frequency (Hz)	Frequency calculated analytically (Hz)	Frequency determined via FEM (Hz)
m	n			
1	1	21.0	21.58	21.51
2	1	33.0	32.7	32.75
1	2	36.3	35.87	35.99
2	2	45.7	43.5	43.33

where ρ is density of the silicone material and the overdot indicates the differentiation with respect to time. The Lagrange equations are used to find the natural frequencies [4]

$$\frac{d}{dt} \left(\frac{dL}{dq_n} \right) - \frac{dL}{dq_n} = Q_n, \quad (12)$$

where $L = T - \Pi$, Q_n are generalized forces and q_n are the generalized coordinates.

In order to model the in-plane pre-stretch, the in-plane displacements u and v are assumed to have the following expressions:

$$u(x, y, t) = \frac{a^F - a}{a} \left(x - \frac{a}{2} \right), \quad v(x, y, t) = \frac{b^F - b}{b} \left(y - \frac{b}{2} \right). \quad (13)$$

As a result of this static deformation (stretching), the rectangular plate with initial dimensions a and b after stretching will have final dimensions a^F and b^F . In our case $a^F = b^F = 0.26$ m.

The deflection is discretized as follows:

$$w(x, y, t) = \sum_{m, n=1}^N w_{n,m}(t) P_n(x) P_m(y), \quad (14)$$

where P_i are orthonormal polynomials satisfying boundary conditions (8):

$$P_1(x) = x^2(x-a)^2 \frac{3\sqrt{70}}{a^{9/2}}, \quad P_2(x) = x^2(x-a)^2 \frac{6\sqrt{770}(x-a/2)}{a^{11/2}}, \dots \quad (15)$$

For convenience we use two-subscript notation $w_{n,m}(t)$ for the generalized coordinates in (14) instead of the single-subscript notation q_n . We observe that there is no coupling between bending and in-plane vibrations, so there is no need to include dynamic terms in expressions (13).

The expression for ε_z should be evaluated in terms of displacements. In order to obtain this expression, we use the incompressibility condition $J=1$, which yields

$$\varepsilon_z = \frac{1}{2 \left((2\varepsilon_x + 1)(2\varepsilon_y + 1) - \varepsilon_{xy}^2 \right)} - \frac{1}{2}. \quad (16)$$

Eqs. (13), (14), (16) are substituted into the Lagrange functional L and the Eq. (12) are obtained. Subsequently, they are linearized, since only the linear vibrations are now of interest. A convergence study has shown that N in (14) should be taken equal to 7 in order to have accurate results. The Rayleigh-Ritz method allows obtaining the approximate analytical expressions for the natural frequencies. These expressions are too long to be given here. By using the analytical expressions for the natural frequencies, the minimization of the error (7) is evaluated with respect to the unknown initial dimensions a and b . The weights p_k are the same as in Section 4.1. There is no need in building the grid of values of a and b , since E_f has now explicit analytical expression. This analytical approach gives practically the same values of the parameters a and b obtained in Section 4.1. The corresponding natural frequencies are listed in Table 2 and they are extremely close to the ones obtained by FE code, but they are also in very good agreement with the experimental ones.

5. Numerical simulations of pressurization tests and comparison to experiments

Numerical simulations of the static deflection of the silicone hyperelastic plate have been carried out by commercial FE code (Abaqus) and a self-developed meshless numerical code based on analytical expressions. In both cases, simulations have been compared to the experimental results for the first loading cycle of the pressurized silicone plate.

5.1. FEM simulations

During the experiments, the pneumatic pressure acting on the plate is increased very slowly, keeping the strain rate low, which eliminates the inertia effects. Thus, this type of experiment can be simulated in FE software as a quasi-static analysis. The pressure is applied on the silicone plate, pre-stretched as in the precedent analysis step (as described in the Section 4.1); loading cycle starts at 0 and finishes at 2.6 PSI. The load is increased linearly throughout the analysis step and the overall number of equally-spaced increments is thirteen thousand. Due to large deformations that occur during the pressurization, the FE analysis implies geometric nonlinearity and plate thickness reduction, as a consequence of the Poisson's effect, in addition to the material nonlinearity described by the Mooney–Rivlin hyperelastic law.

We further compare the results obtained by the FE analysis with the experimental data from the first loading cycle. Confrontation of the results is depicted in Fig. 11. Deviation from experimental results shows that the model tends to be more flexible compared to the actual plate, but results agree quite well.

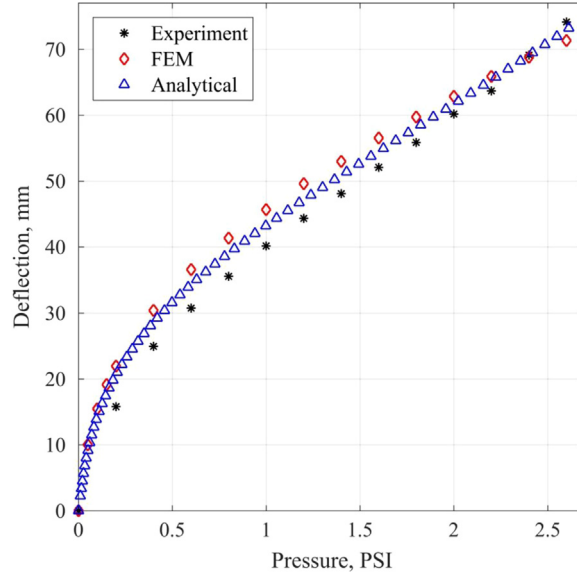


Fig. 11. Static deflection of the silicone plate measured at the center as function of the pressure. *, experimental result (first loading cycle); $\diamond$, FEM results; Δ , analytical results.

5.2. Computations by meshless approach

The finite element method requires significant number of degrees of freedom. At the same time, it is possible to simulate the bending of physically nonlinear plate using much a smaller number of DOFs [3,6]. To this end, we discretize equations (9) by using the following expansions for the displacements:

$$w(x, y) = \sum_{n, m=1}^N w_{2n+1, 2m+1} P_{2n+1}(x) P_{2m+1}(y), \quad (17a)$$

$$u(x, y) = \frac{a^F - a}{a} \left(x - \frac{a}{2} \right) + \sum_{n, m=1}^N u_{2n, 2m+1} R_{2n}(x) R_{2m+1}(y), \quad (17b)$$

$$v(x, y) = \frac{b^F - b}{b} \left(y - \frac{b}{2} \right) + \sum_{n, m=1}^N v_{2n+1, 2m} R_{2n+1}(x) R_{2m}(y), \quad (17c)$$

where R_i are orthonormal polynomials satisfying the boundary conditions (8), the first of them have the forms:

$$R_1(x) = x(x-a) \frac{\sqrt{30}}{a^{5/2}}, \quad R_2(x) = -x(x-a) \frac{\sqrt{210}(a-2x)}{a^2}, \dots \quad (18)$$

The expressions (15) and (18) are substituted in (17a–c) and subsequently into equations (9a–c). However, we see that the expression (16) is non-polynomial in strains, which significantly complicates the investigation of the plate behavior; in fact, it is impossible to carry out analytical spatial integration of the strain energy density (SED).

To overcome this difficulty we employ the local models method [6], which consists in the consecutive expansion of the SED into truncated series in the strain tensor components. This allows to deal with approximate expressions of the SED in polynomial form and, hence, with polynomial algebraic equations (12).

Since the structure undergoes large deformations, we use the expression for pressure as a follower load distribution [20], i.e. it is displacement dependent. The virtual work for the pressure acting on rectangular plate is [20]:

$$W = P \int_0^a \int_0^b (w(1 + u_x + v_y + u_x v_y - u_y v_x) + u(-w_x - w_x v_y - w_y v_x) + v(-w_y - w_y u_x - w_x u_y)) dx dy. \quad (19)$$

The convergence study has shown that sufficient accuracy is achieved by using four terms in each of expressions (17a–c), which results in 12 degrees of freedom. The computed pressure–deflection curve is shown in Fig. 11 and is very close to the FE results. A good agreement with the experimental results is observed.

6. Conclusions

The proposed model satisfactorily predicts the hyperelastic behavior of a silicone rubber plate, in case of linear dynamics and nonlinear static loading. As polymeric structures often require preload during installation, tuning and fitting procedures are required to include differential stiffness into the model. Small amplitude vibrations are mostly influenced by the amount of preload, while the nonlinear material behavior has limited influence, as the amount of dynamic deformation is modest. Modal damping values estimated by experimental modal analysis are relatively large, especially if compared to similar structures composed of stiffer materials (e.g. metal). The correspondence between in-plane preload and forced vibration response is described correctly and in a similar way by the proposed analytical and finite element models. A simplified traction test is sufficient to characterize the material in a wide range of deformations, according to a two-parameter Mooney–Rivlin model for incompressible materials. As material identification and preload estimate are correctly performed, the finite element and the reduced order model predict with impressive accuracy experimental results up to large deformation values.

The experimental activity, however, requires particular care, as hyperelasticity does not define completely the material behavior of the plate. In case of very large deformations and loading cycles, time-dependent phenomena appear, which should be identified appropriately in future studies. The incompressibility of silicone also makes very difficult the implementation of ideal boundary conditions. To the authors' best knowledge, this is the first experimental study of hyperelastic square plates under forced vibration and large amplitude pressure deflection. However it is desirable, as a future development, the extension of the experimental study to large-amplitude vibrations. Aerostatic pressure was used here to reach large static deformations of the plate.

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